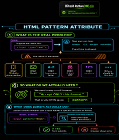


Hitesh Rathore(HR)

HTML PATTERN ATTRIBUTE

1 WHAT IS THE REAL PROBLEM?

Suppose we create this:

`<input type="text">`

Now user can type:

Hitesh 123 abc@# hello999

Everything is allowed.

But what if we want:

SO WHAT DO WE ACTUALLY NEED ?

We need a way to tell browser:

"Accept ONLY this format."

That is why HTML gives: pattern

WHAT DOES pattern ACTUALLY DO?

pattern checks whether user's input follows a specific structure or format.

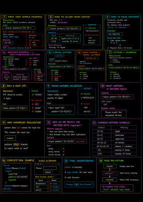
BASIC SYNTAX

`<input pattern="RULE">`

Browser internally checks: Does user input match this RULE?

If YES: form submits

If NO: browser shows error



1 FIRST VERY SIMPLE EXAMPLE

2 HOW TO ALLOW MANY DIGITS?

3 HOW TO READ PATTERN?

4 ANOTHER EXAMPLE → ONLY LOWERCASE LETTERS

5 CAPITAL LETTERS

6 BOTH CAPITAL + SMALL LETTERS

7 LETTERS + NUMBERS

8 ONLY 6 DIGIT OTP

Requirement

OTP should be exactly 6 digits

Code

`<input pattern="[0-9]{6}">`

Allowed: ✓ 123456

Not Allowed: ✗ 123 ✗ abcdef ✗ 12ab56

9 PHONE NUMBER VALIDATION

Requirement

Indian mobile number: exactly 10 digits

Code

`<input type="tel" pattern="[0-9]{10}">`

OUTPUT

✓ Allowed ✓ 9876543210

✗ Invalid 999 abc123

10 WHAT HAPPENS IF PATTERN FAILS?

Suppose: `<input pattern="[0-9]{10}">`

User types: 123

Browser shows: Please match the requested format.

11 VERY IMPORTANT REALIZATION

pattern does NOT create the input box.

This creates the input box: `<input>`

pattern ONLY checks: Is input valid or not?

12 WHY DO WE MOSTLY USE PATTERN WITH required ?

Without required:

- User can leave field empty.
- And browser may still allow submission.

Example

`<input pattern="[0-9]{10}" required>`

Now: empty ✗ wrong format ✗ correct format ✓

13 COMMON PATTERN SYMBOLS

Pattern	Meaning
[0-9]	digits
[a-z]	lowercase
[A-Z]	uppercase
[A-Za-z]	all letters
[A-Za-z0-9]	letters + numbers
{5}	exactly 5 times
{10}	exactly 10 times

14 COMPLETE REAL EXAMPLE

15 FINAL UNDERSTANDING

pattern is basically: A rule checker for user input.

It tells browser: "Accept ONLY this format."

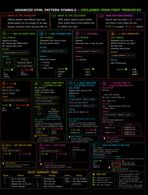
16 FINAL BIG PICTURE

`<input>` Creates input box

pattern="" Adds format checking

required Makes field compulsory

So together they create: Smart validated input fields



ADVANCED HTML PATTERN SYMBOLS – EXPLAINED FROM FIRST PRINCIPLES

WHAT IS THE PROBLEM?

Different websites need different input rules. Simple patterns are not enough for all cases. Example: Username, Email, Password, PIN, etc.

WHAT IS THE SOLUTION?

HTML pattern supports special symbols. These symbols help us write powerful rules to validate user input.

HOW PATTERN WORKS?

Browser reads the pattern LEFT → RIGHT and checks if input follows the rule. If YES accepted If NO rejected

1 + → ONE OR MORE TIMES

2 * → ZERO OR MORE TIMES

3 ? → OPTIONAL

4 . → ANY CHARACTER

5 | → OR

6 ^ → MUST START WITH (^ is OPTIONAL in HTML)

7 $ → MUST END WITH ($ is OPTIONAL in HTML)

8 \. → ACTUAL DOT (.) (Single backslash before dot)

9 [abc] → ANY ONE OF THESE

10 [^abc] → NOT THESE

11 .* → ZERO OR MORE OF ANYTHING

QUICK SUMMARY TABLE

Symbol	Meaning	Symbol	Meaning	Symbol	Meaning
+	One or more times	|	OR (either/or)	[abc]	Any one of these
*	Zero or more times	^	Must start with	[^abc]	Not these (except)
?	Optional (may or may not)	$	Must end with	\.	Actual dot
.	Any one character	.*	Zero or more of anything		

PRO TIPS

1. Understand the rule
2. Break into parts
3. Choose correct symbol
4. Test with examples

Pattern Validation Practice

1. Allow ONLY lowercase letters

2. Allow ONLY uppercase letters

3. Allow ONLY numbers

4. Allow exactly 5 digits

5. Allow lowercase and uppercase letters only

6. Only vowels

7. Allow ONLY Gmail addresses

8. Only binary digits

9. Username should start with a capital letter

10. Allow exactly 3 lowercase letters followed by 3 digits

11. Indian phone number

12. Starts with letter ends with digit

13. Date Validation (DD-MM-YYYY)

14. Hexadecimal Color Code Validation

15. Simple URL validation

Q) Check whether the input is a 7-character palindrome or not.

Capturing Groups in Regex — Complete Notes

1. What is a Capturing Group?

2. Why Do We Need Capturing Groups?

3. Basic Example

4. Group Numbering

5. Backreferences

6. Important Example

7. Main HTML Pattern Example

Practice Questions on Capturing Groups

Q1. 7-Character Palindrome

Q2. Two Same Digits Together Anywhere

Q3. Word Repeated Twice

Q4. Starts and Ends with Same Digit and Only Letters in Middle

Final Core Concept

() vs (?:)

Capturing Group vs Non-Capturing Group

1. Capturing Group ()

Name

Syntax

Purpose

Example

Meaning

Matches

Captured Group

Step-by-Step Breakdown

2. Non-Capturing Group (?:)

Name

Syntax

Purpose

Example

Meaning

Matches

Captured Group

Step-by-Step Breakdown

Main Difference

Why is (?:) Used?

Example 1

Capturing Version

Non-Capturing Version

Example 2

Capturing Version

Non-Capturing Version

Example 3

Capturing Version

Non-Capturing Version

In Short

Q1. String should NOT contain repeated adjacent letters , write pattern.

Q2. Password Validation

- At least 1 uppercase letter required
- At least 1 lowercase letter required
- At least 1 number required
- Minimum length should be 8

Lookahead in Regex — Complete Notes

1. What is Lookahead?

2. Positive Lookahead

Example 1

Important Point

Example 2

3. Negative Lookahead

Example 1

Example 2

4. Difference Between Positive and Negative Lookahead

5. Hard Questions on Negative Lookahead

Q1. String should NOT contain two consecutive digits

Q2. String should NOT start with a digit

Q3. String should NOT end with "ing"

Q4. String should NOT contain repeated adjacent letters

Q5. String should NOT contain palindrome-like pattern ABA

6. Good Questions on Positive Lookahead

Q1. Match a letter only if followed by a digit

Q2. Match a digit only if followed by two letters

Q3. Password Validation

Q4. String must contain both @ and

Q5. String must contain a digit and end with a letter

7. Final Important Points

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1. Exactly 10 chars but not all digits

2. Ends with jpg or png

3. Must contain 3 uppercase letters

4. Only letters but not hello

5. Contains exactly one digit